

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.IT Semester V)
Data Visualization with Power BI and Tableau

Practical – II

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Class: SYIT	Batch: 1
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AIM: Working with Data Sources and Data Preparation in Tableau.

1. Connect to different data sources (Excel, CSV, Database)
2. Create and use data extracts vs live connections
3. Manage metadata (rename, hide, change data types)
4. Perform joins (inner, left, right) on multiple tables
5. Apply data blending between two sources
6. Implement different types of filters (dimension, measure, date)
7. Optimize performance using extracts and filtering

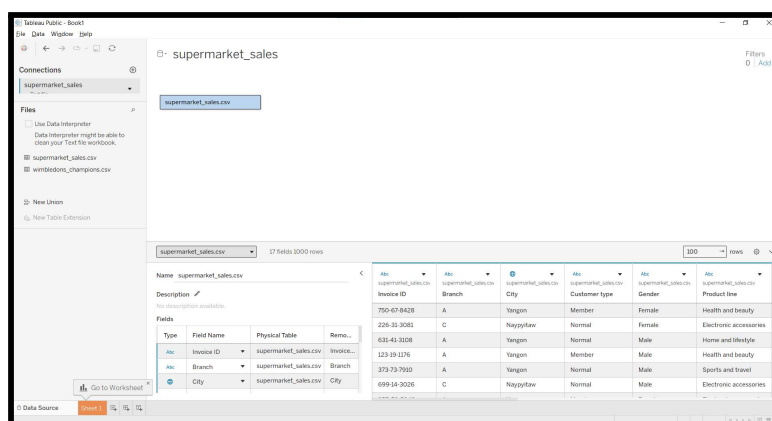
STEPS:

A. Connecting to Different Data Sources

Step 1 – Connect to the .CSV File

1. Open Tableau Desktop.
2. Under the Connect section, click on the text file.
3. Browse to the downloaded file supermarket_sales.csv.
4. Select the file and click Open.
5. Wait for Tableau to load the dataset into the Data Source page.

Screenshot 1: Data Source page after loading supermarket_sales.csv.

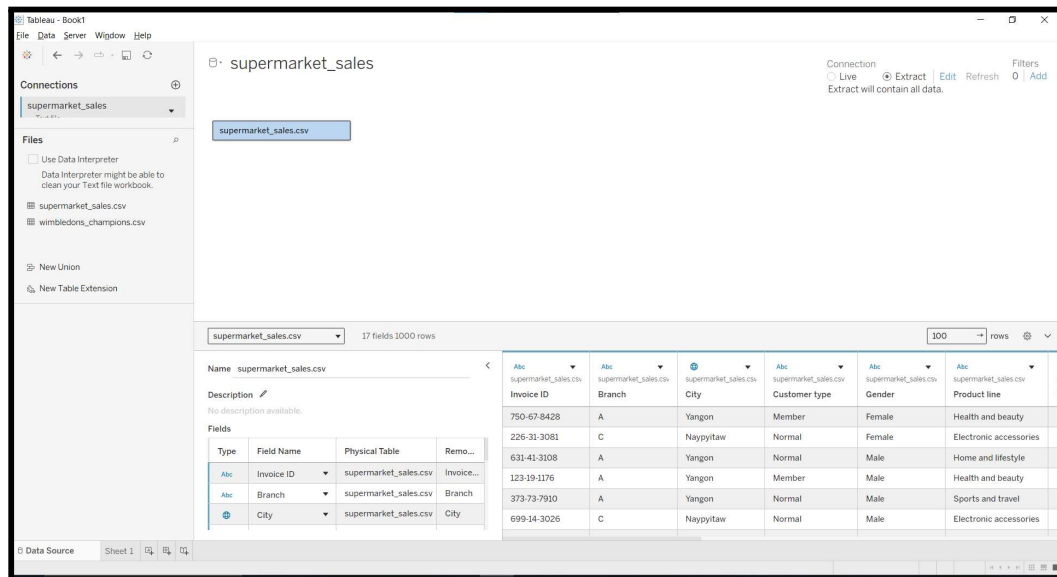


B. Creating Data Extracts vs. Live Connections

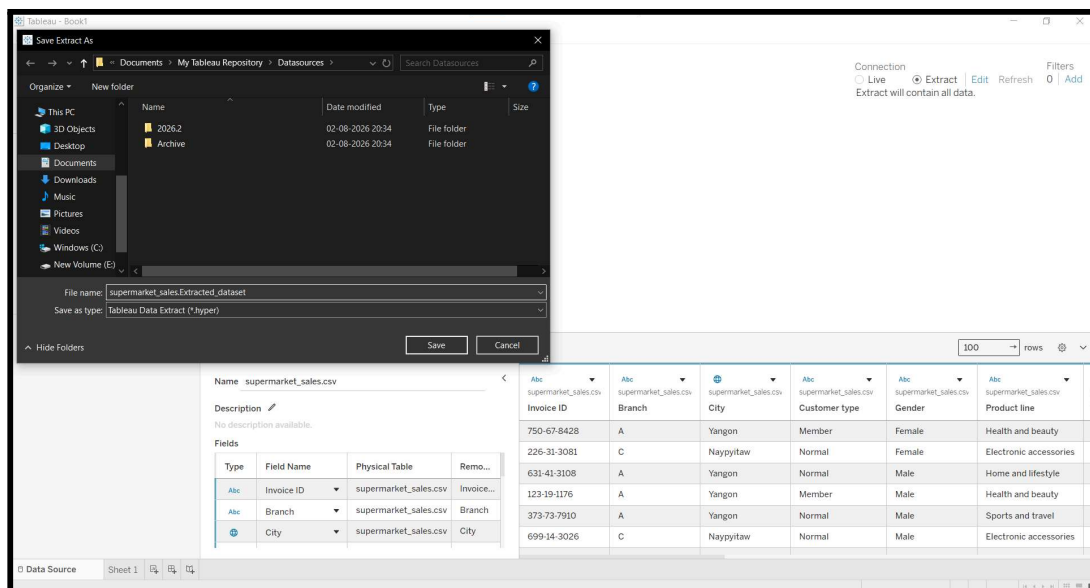
Step 2 – Create an Extract Connection

1. After the dataset is loaded in the **Data Source** tab, navigate to the **top-right corner** of the screen.
2. In the **Connection** section, switch the connection mode from **Live** to **Extract**.
3. Click the **Sheet 1** tab located at the **bottom-left** of the Tableau window.
4. When the **Save Extract** dialog box appears, enter **Ecommerce_Extract** as the file name.
5. Click **Save** to create and store the extract file, then continue working in the worksheet.

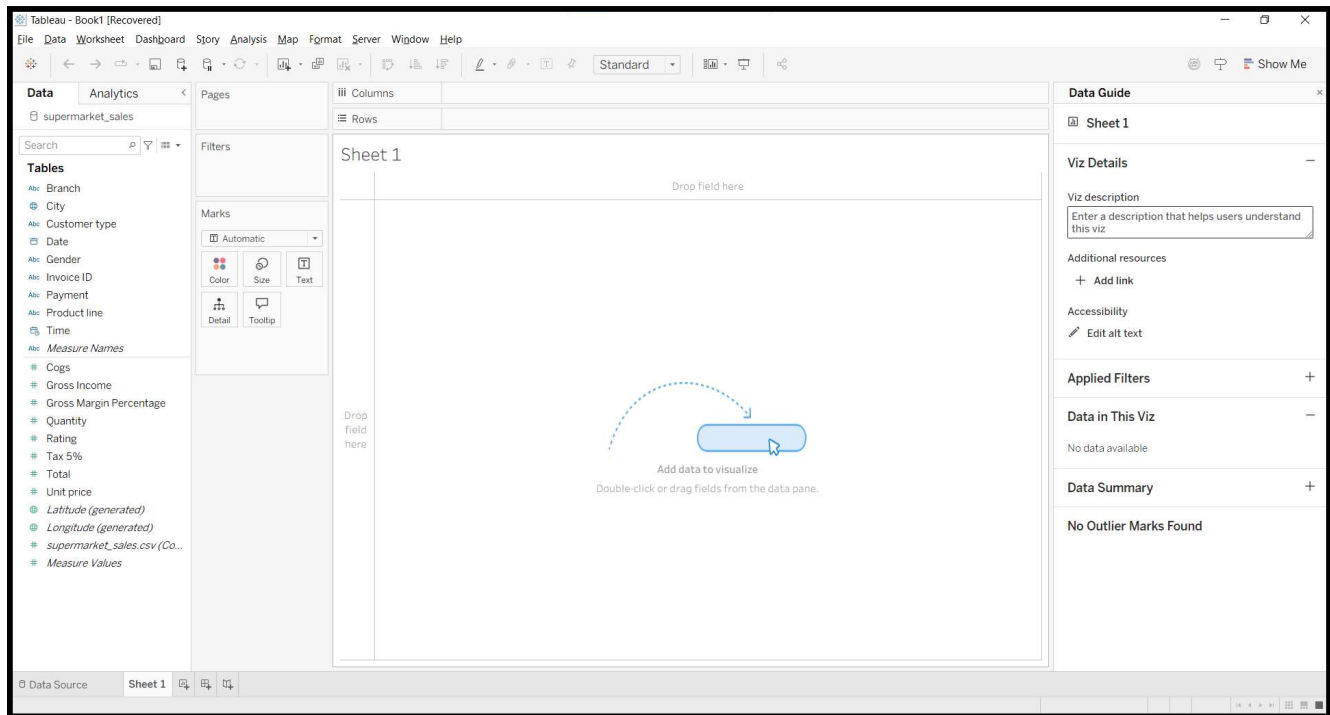
Screenshot 2: Connection mode changed from Live to Extract on the Data Source page.



Screenshot 3: Save Extract dialog box with the file name Ecommerce_Extract entered before saving.



Screenshot 4: Worksheet (Sheet 1) opened successfully after saving the extract file.

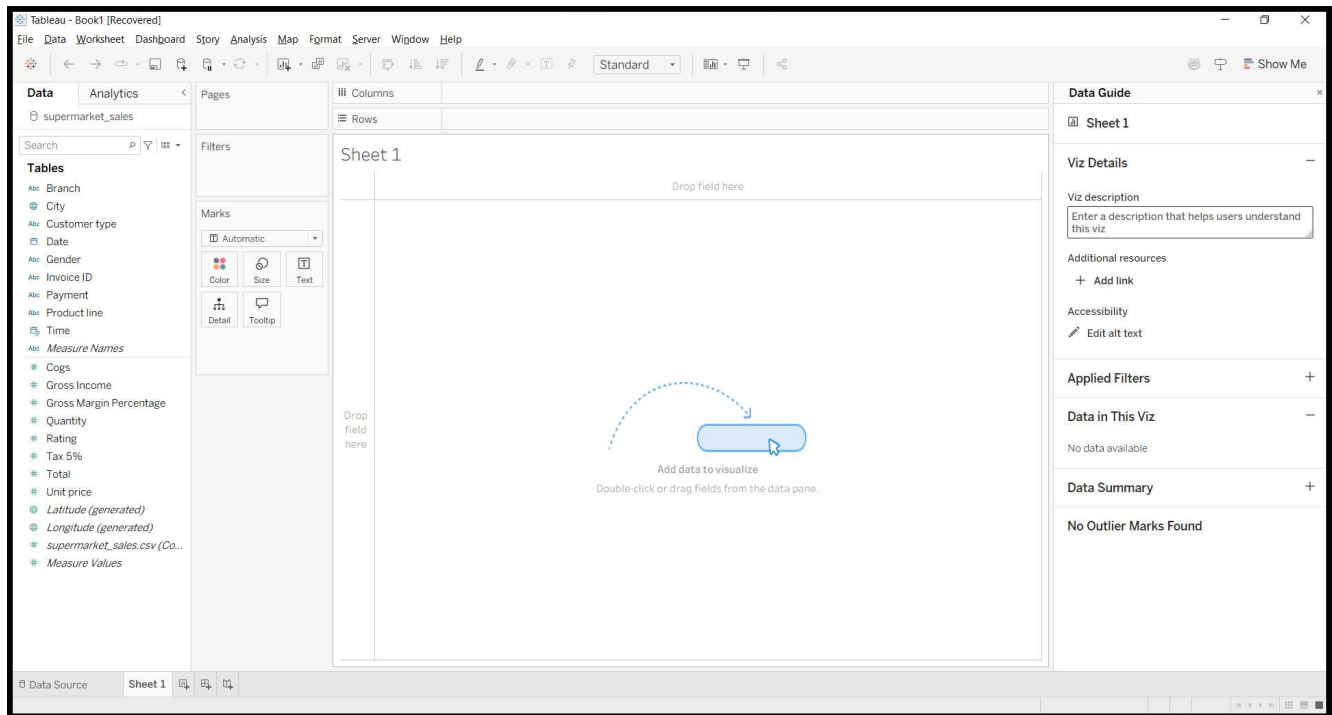


C. Managing Operational Metadata.

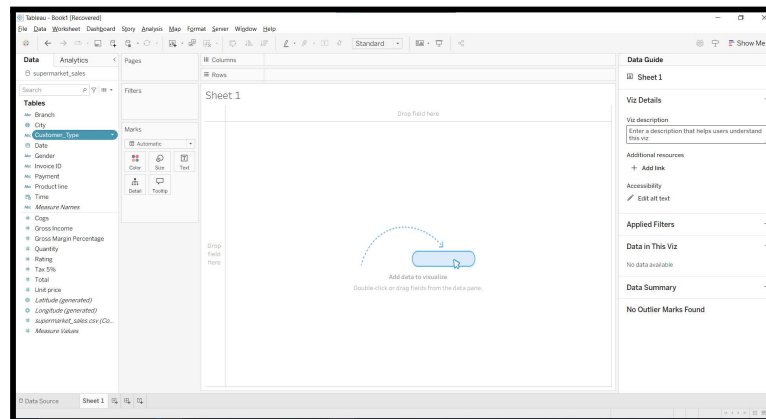
Step 3 – Rename, Hide, and Change Data Type of Fields

1. Open Sheet 1 and view the Data Pane on the left side of the workspace.
2. Locate the field Customer type, right-click it, select Rename, enter Customer_Type, and press Enter.
3. Locate the field Invoice ID, right-click it, and select Hide to remove it from the Data Pane.
4. Find the Date field and click the data type icon beside it.
5. If the field is detected as Text (Abc), change it to Date from the dropdown menu to ensure proper date recognition.

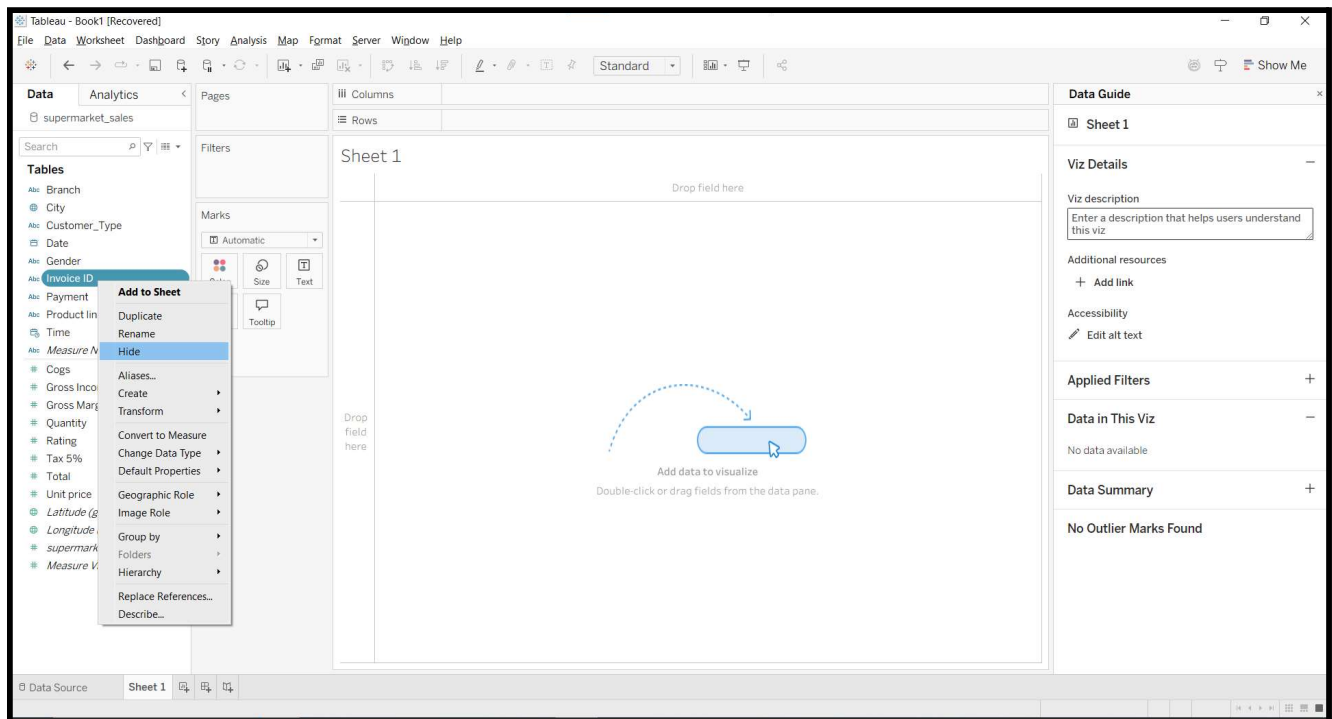
Screenshot 5: Data Pane in Sheet 1 displaying all fields from the supermarket_sales.csv dataset.



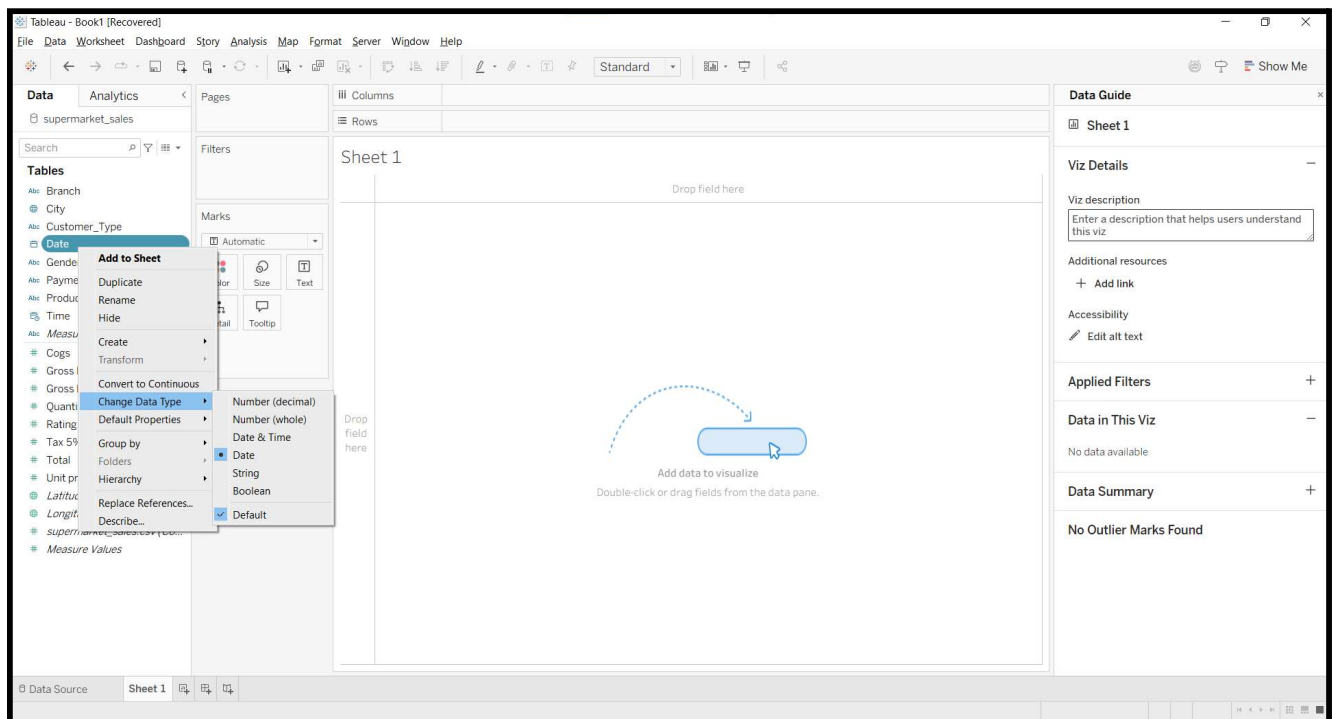
Screenshot 6: The Customer type field being renamed to Customer_Type.



Screenshot 7: The Invoice ID field hidden from the Data Pane.



Screenshot 8: The Date field data type changed to Date for correct date handling in Tableau.



D. Performing Relational Joins on Multiple Tables.

Step 4 – Join Orders and Order Payments Tables

1. Click the **Data Source** tab at the bottom-left corner to return to the Data Source page.
2. Locate the **olist_orders_dataset.csv** table on the canvas.
3. Double-click the **olist_orders_dataset.csv** table to enter the **Physical Layer**.
4. From the left pane, drag **olist_order_payments_dataset.csv** next to the **olist_orders_dataset.csv** table to create a join.
5. Click the **Join** (Venn Diagram) icon between the two tables.
6. Select **order_id** as the matching field for both the left and right tables. (Tableau may select it automatically.)
7. Test the join types:
 - a. Select **Inner Join** to display only matching order and payment records.
 - b. Select **Left Join** to display all orders and their matching payment details, leaving unmatched payment fields blank.
8. Keep the join type set to **Left Join** and continue with the analysis.

Screenshot 9: Data Source page showing the olist_orders_dataset.csv table loaded.

The screenshot shows the Tableau Desktop interface with the Data Source page for the **olist_orders_dataset.csv** table. The left pane displays the **Connections** and **Files** sections. The central canvas shows the table name **olist_orders_dataset.csv**. The bottom pane displays the table's fields and a preview of the data.

Type	Field Name	Physical Table	Rem...
ABC	Order Id	olist_orders_dataset...	order_id
ABC	Customer Id	olist_orders_dataset...	custo...
ABC	Order Status	olist_orders_dataset...	order_...

The data preview shows the following columns: Order Id, Customer Id, Order Status, Order Purchase Timestamp, Order Approved At, and Order. The data is as follows:

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approved At	Order
e481f51cdbc5467...	9ef432eb625129...	delivered	02-10-2017 10:56:33	02-10-2017 11:07:15	04-10-
53cdb2fc8bc7dce...	b0830fb4747a6c...	delivered	24-07-2018 20:41:37	26-07-2018 03:24:27	26-07-
47770eb9100c2d...	41ce2a54c0b03b...	delivered	08-08-2018 08:38:49	08-08-2018 08:55:23	08-08-
949d5b44dbf5de...	f88197465ea792...	delivered	18-11-2017 19:28:06	18-11-2017 19:45:59	22-11-
ad21c59c0840e6...	8ab97904e6daaa...	delivered	13-02-2018 21:18:39	13-02-2018 22:20:29	14-02-
a4591c265e18cbl...	503740e9ca751c...	delivered	09-07-2017 21:57:05	09-07-2017 22:10:13	11-07-

Screenshot 10: Physical Layer displaying the *olist_orders_dataset.csv* and *olist_order_payments_dataset.csv* tables.

Tableau - Book1 [Recovered]

File Data Server Window Help

Connections: olist_orders_dataset

Files: Use Data Interpreter, olist_order_payments_dataset.csv, olist_order_reviews_dataset.csv, olist_orders_dataset.csv, New Union, New Table Extension

olist_orders_dataset+ Connection: Live, Extract, Filters: 0 | Add

olist_orders_dataset.csv is made of 2 tables. ①

olist_orders_dataset.csv

13 fields 103886 rows

Name: olist_orders_dataset.csv

Description: No description available.

Type	Field Name	Physical Table	Rem...
Abc	Order Id	olist_orders_dataset...	order_id
Abc	Customer Id	olist_orders_dataset...	custo...
Abc	Order Status	olist_orders_dataset...	order_...

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approved At
75729cd39e546e...	aa3e69930a8912...	delivered	09-02-2017 19:33:07	09-02-2017 19:43:13
1d2a2c1119967c4...	db174a68814fb58...	delivered	24-11-2017 19:03:03	24-11-2017 22:32:42
be96af68ea7acf4...	c4b925691c0532...	delivered	21-01-2018 11:46:11	22-01-2018 13:47:47
732c9ceb978695...	ee24608e86f735...	delivered	05-01-2018 22:10:39	06-01-2018 22:07:25
64c8df7b489081...	f33a18deacf7ae3...	delivered	09-07-2017 13:44:10	11-07-2017 03:55:46
4259e68b0021b...	bbf16d44cf6f0d9...	delivered	01-08-2017 16:51:20	01-08-2017 17:05:18

Screenshot 11: Join configuration with *order_id* selected as the matching field.

Tableau - Book1 [Recovered]

File Data Server Window Help

Connections: olist_orders_dataset

Files: Use Data Interpreter, olist_order_payments_dataset.csv, olist_order_reviews_dataset.csv, olist_orders_dataset.csv, New Union, New Table Extension

olist_orders_dataset+ Connection: Live, Extract, Filters: 0 | Add

olist_orders_dataset.csv is made of 2 tables. ①

olist_orders_dataset.csv

13 fields 103886 rows

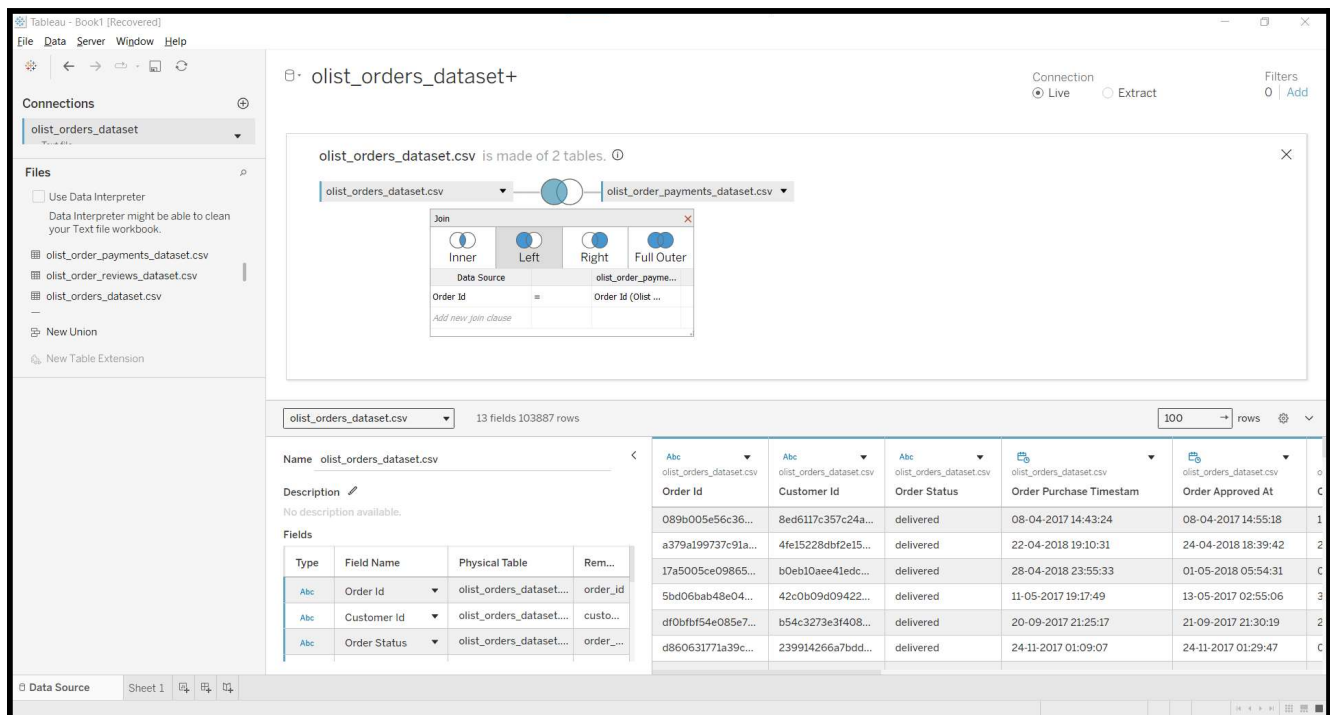
Name: olist_orders_dataset.csv

Description: No description available.

Type	Field Name	Physical Table	Rem...
Abc	Order Id	olist_orders_dataset...	order_id
Abc	Customer Id	olist_orders_dataset...	custo...
Abc	Order Status	olist_orders_dataset...	order_...

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approved At
75729cd39e546e...	aa3e69930a8912...	delivered	09-02-2017 19:33:07	09-02-2017 19:43:13
1d2a2c1119967c4...	db174a68814fb58...	delivered	24-11-2017 19:03:03	24-11-2017 22:32:42
be96af68ea7acf4...	c4b925691c0532...	delivered	21-01-2018 11:46:11	22-01-2018 13:47:47
732c9ceb978695...	ee24608e86f735...	delivered	05-01-2018 22:10:39	06-01-2018 22:07:25
64c8df7b489081...	f33a18deacf7ae3...	delivered	09-07-2017 13:44:10	11-07-2017 03:55:46
4259e68b0021b...	bbf16d44cf6f0d9...	delivered	01-08-2017 16:51:20	01-08-2017 17:05:18

Screenshot 12: Left Join successfully applied between the Orders and Order Payments tables.

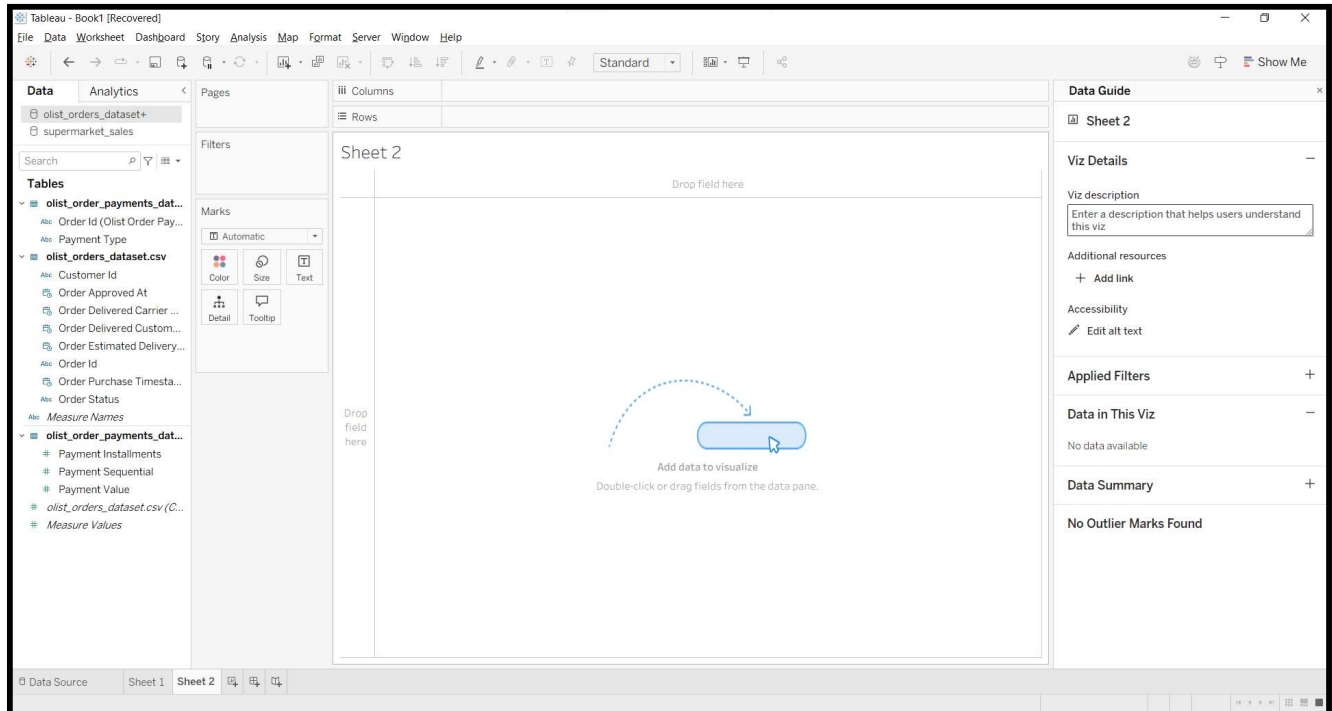


E. Applying Data Blending Between Independent Sources.

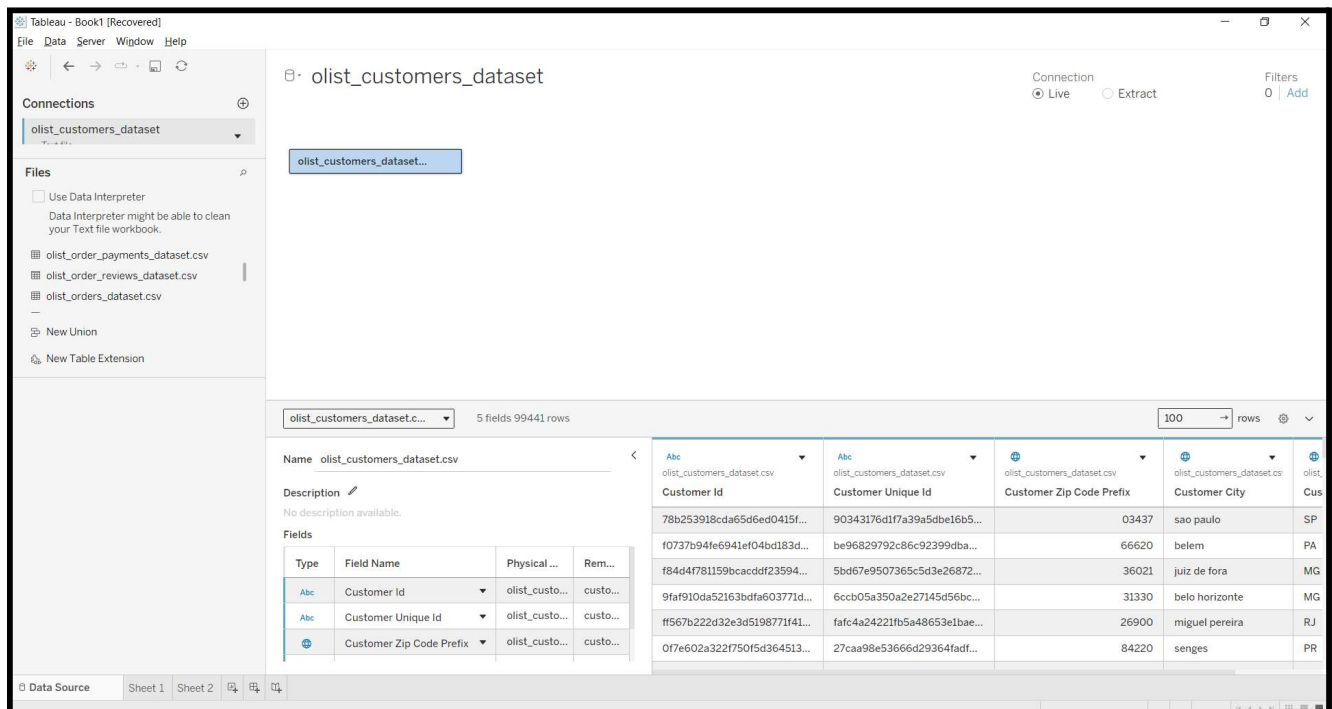
Step 5 – Apply Data Blending Between Independent Data Sources

1. Create a new worksheet by clicking the **New Worksheet (+)** icon to open **Sheet 2**.
2. In the **Data Pane**, select the **Orders** data source and drag the **Status** field to the **Rows** shelf.
3. Go to **Data** → **New Data Source** and add the **olist_customers_dataset.csv** file.
4. Return to **Sheet 2**, then click **Data** → **Edit Blend Relationships**.
5. Set **Orders** as the **Primary** data source and **Customers** as the **Secondary** data source.
6. Change the relationship from **Automatic** to **Custom**, then map **customer_id** from both datasets and click **OK**.
7. If **customer_id** is hidden, return to the **Data Source**, switch the connection to **Live**, go back to **Sheet 2**, select **Show Hidden Fields**, and unhide **customer_id**.
8. In the **Customers** data source, click the chain icon next to **customer_id** to activate the blend.
9. Drag **customer_state** from the **Customers** data source to the **Columns** shelf.

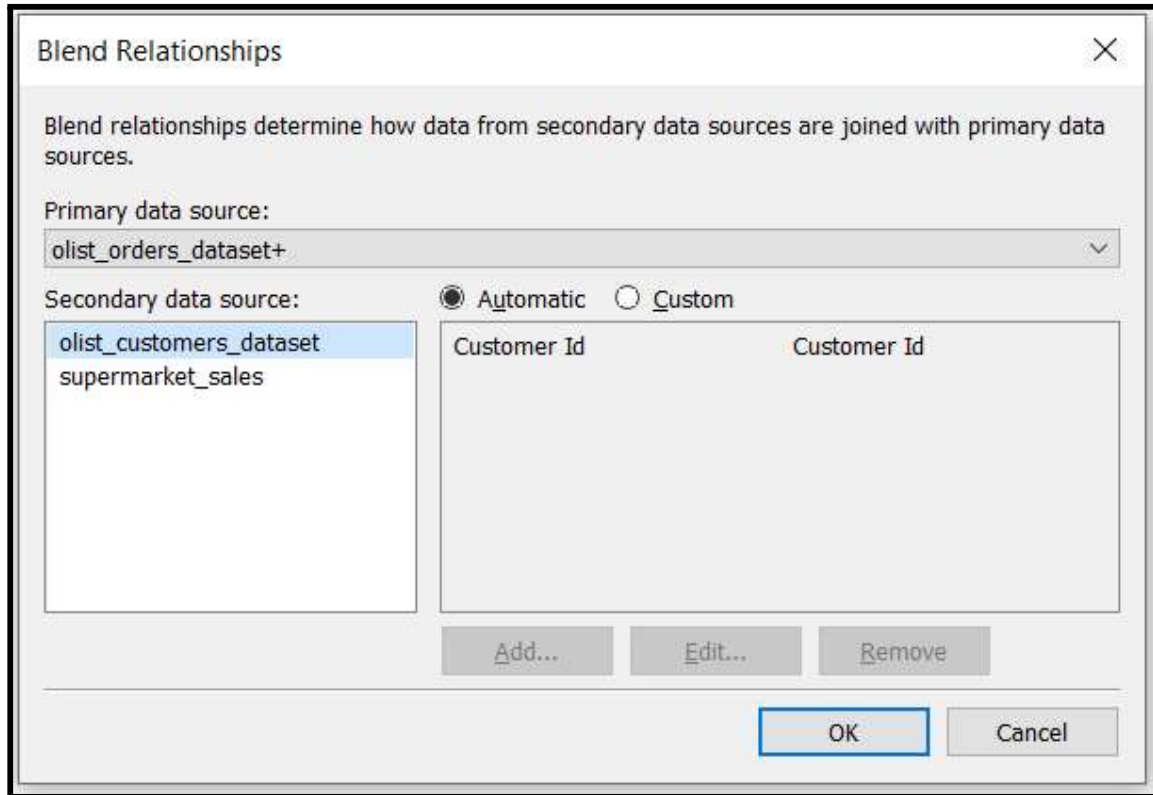
Screenshot 13: New worksheet (Sheet 2) created with the Orders data source selected.



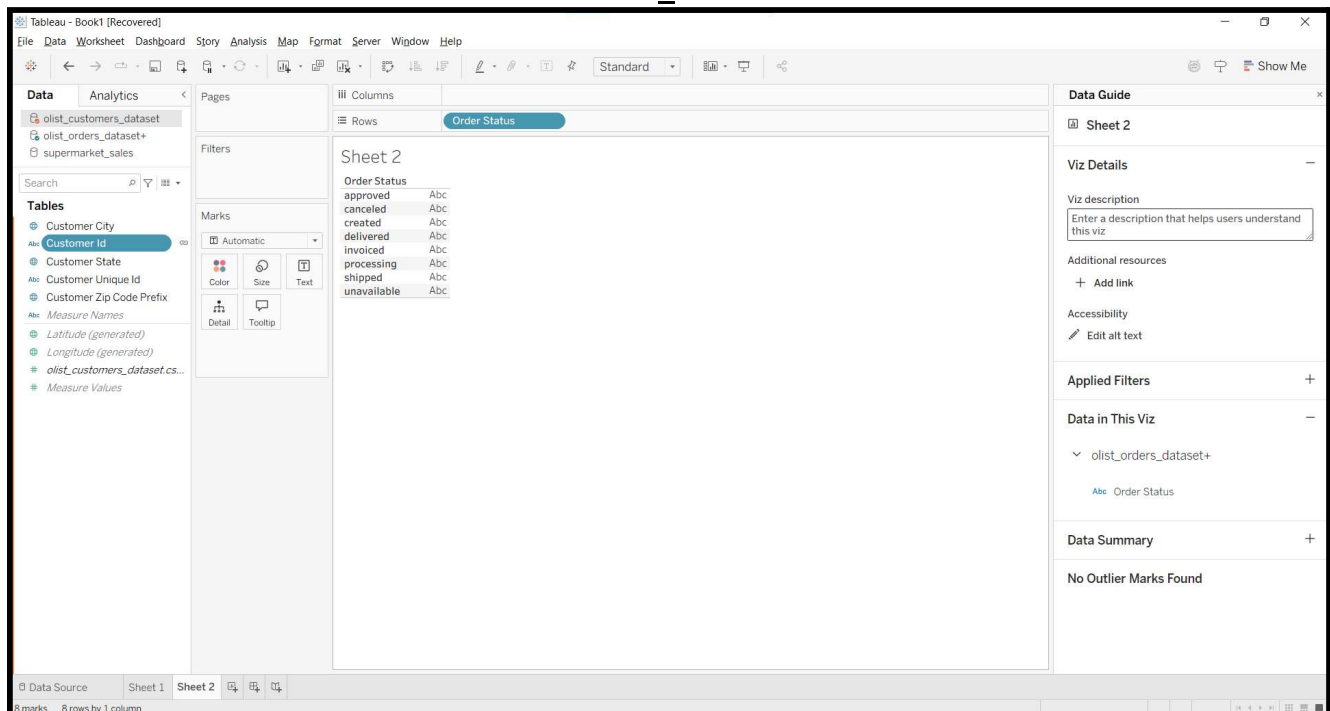
Screenshot 14: Customers dataset added as a new data source.



Screenshot 15: Edit Blend Relationships configured using customer_id.



Screenshot 16: Chain link activated on customer_id to establish the data blend.



Screenshot 17: Blended view displaying Status and customer_state on the worksheet.

Tableau - Book1 [Recovered]

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Columns: Customer State

Rows: Order Status

Sheet 2

Order Status	AC	AL	AM	AP	BA	CE	DF	ES	GO	MA	MG	MS	MT	PA	PB	PE	PI	PR	RJ	RN	RO	RR	RS
approved																							
canceled																							
created																							
delivered	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc
invoiced	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc
processi..	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc
shipped	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc
unavaila..	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc

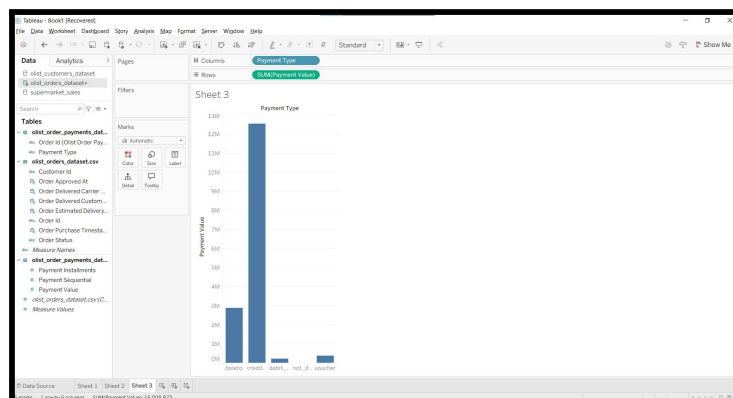
149 marks 8 rows by 27 columns

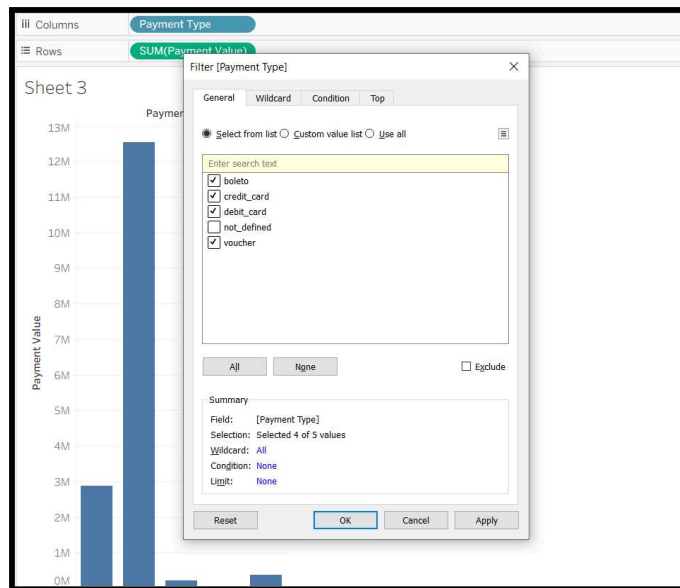
F. Implementing Multi-Tiered Filters.

Step 6 – Apply Dimension, Measure, and Date Filters

1. Open a new worksheet and drag **Payment Value** to the **Rows** shelf and **Payment Type** to the **Columns** shelf.
2. Drag **Payment Type** to the **Filters** shelf, uncheck **not_defined**, select **credit_card** and **boleto**, then click **OK**.
3. Drag **Payment Value** to the **Filters** shelf, choose **All Values**, click **Next**, select **At Least**, enter **50**, and click **OK**.
4. Drag **Order Purchase Timestamp** to the **Filters** shelf, choose **Years** or **Relative Date**, select the required date range, and click **OK**.

Screenshot 18: Payment Value and Payment Type added to the worksheet.

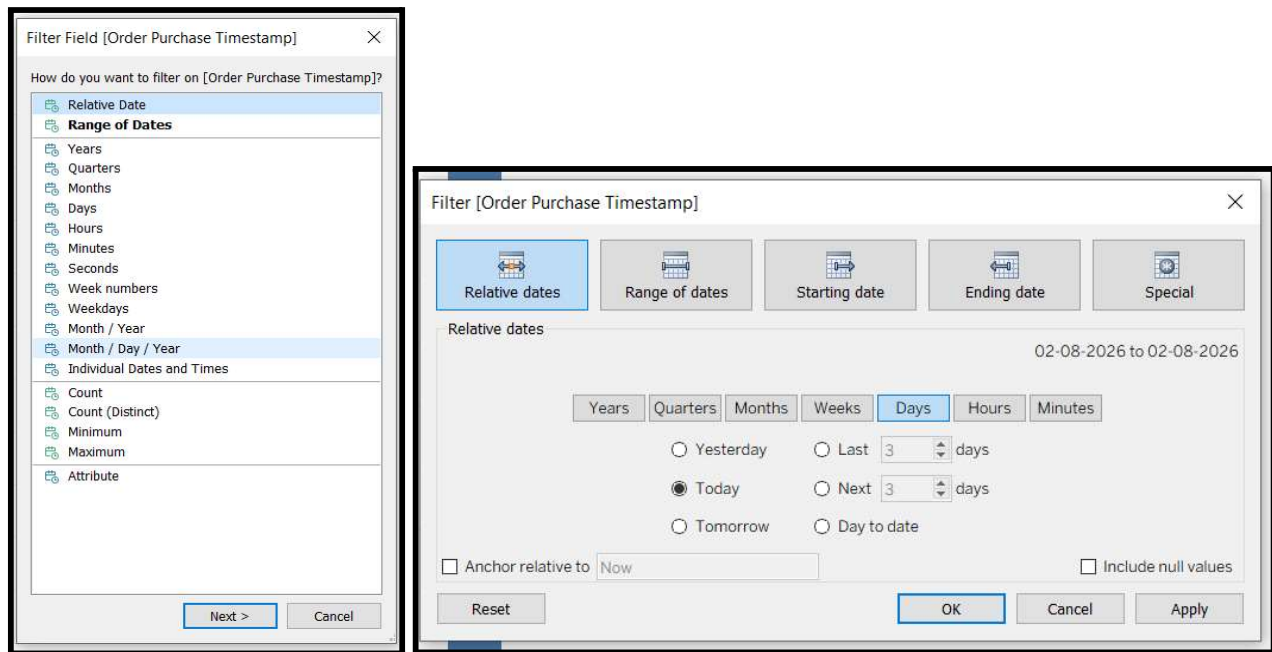


Screenshot 19: *Payment Type* filter applied with *credit_card* and *boleto* selected.Screenshot 20: *Payment Value* filter set to show values greater than or equal to 50.

The first screenshot shows the 'Filter Field [Payment Value]' dialog box. The question 'How do you want to filter on [Payment Value]?' is at the top. A list of filter options is shown, with 'All values' selected. Other options include Sum, Average, Median, Count, Count (Distinct), Minimum, Maximum, Standard deviation, Standard deviation (Population), Variance, Variance (Population), and Attribute. 'Next >' and 'Cancel' buttons are at the bottom.

The second screenshot shows the 'Filter [Payment Value]' dialog box. The 'At least' filter type is selected. The 'At least' section shows a range from 50 to 13,664.08. The 'Show:' dropdown is set to 'Only Relevant Values'. There is an unchecked checkbox for 'Include Null Values'. 'Reset', 'OK', 'Cancel', and 'Apply' buttons are at the bottom.

Screenshot 21: Date filter applied using the Order Purchase Timestamp field.

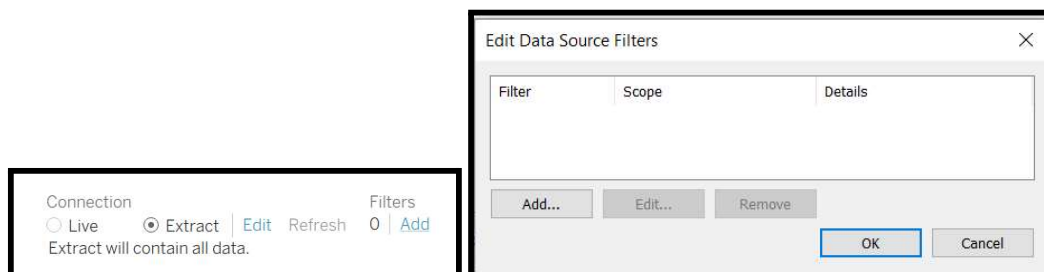


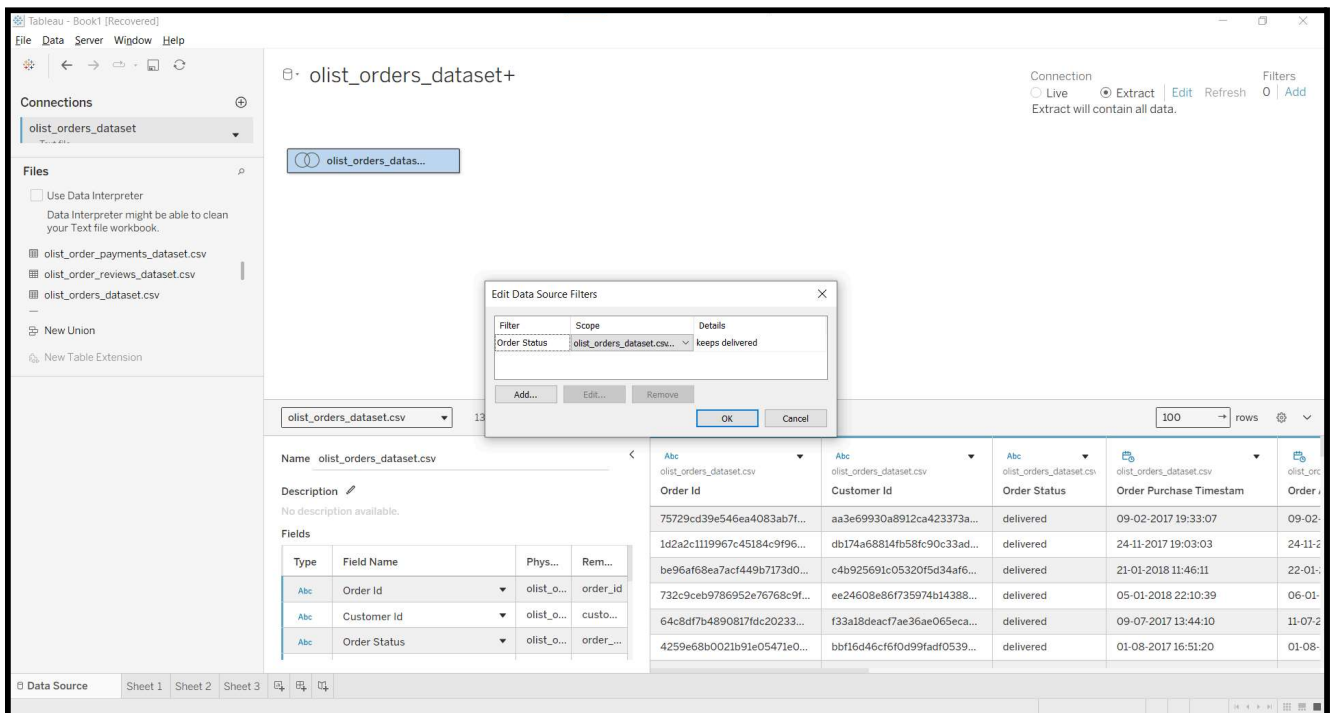
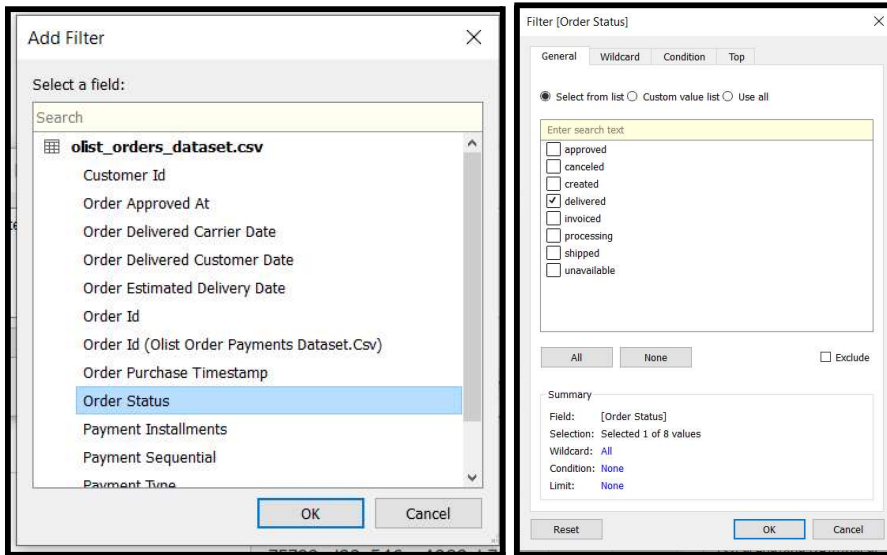
G. Optimizing Dashboard Performance.

Step 7 – Apply Extract and Context Filters

1. Go to the **Data Source** page and click **Edit** under the **Extract** option.
2. Click **Add**, select the **Order Status** field, choose **delivered**, and click **OK** to create an extract containing only delivered orders.
3. Return to the worksheet and locate the **Filters** shelf.
4. Right-click any active filter (such as **Payment Type** or **Order Status**) and select **Add to Context**. The filter will turn **gray**, indicating it has become a context filter.

Screenshot 22: Extract Filter configured to include only delivered orders.





Screenshot 23: Context Filter applied, showing the filter pill in gray on the Filters shelf.

